

A self-contained proof of a polynomial-scale gap between the chromatic and cochromatic numbers of a random graph

Candidate all-n solution to Erdős Problem 625; not externally peer reviewed

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Contents

Abstract	2
1. Notation and elementary facts	2
2. The complete independence-number phase	3
3. Continuous profile roots	4
4. A valid unrestricted lower location for χ	7
5. The four-size signed first-moment advantage	8
6. Exact signed second-moment representation	11
7. Exact partial diagonals	12
8. Canonical high cells and dense endpoint transportation	16
9. All residual local and cycle attachments	21
10. Rare-event amplification	24
11. Completion of the proof	26
12. Dependency and provenance note	26
13. References	27

Abstract

For a graph G , let $\chi(G)$ be its chromatic number and let $\zeta(G)$ be the least number of parts in a vertex partition in which every part induces either an empty graph or a complete graph. We prove that, for $G_n \sim G(n, 1/2)$,

$$\Pr \left\{ \chi(G_n) - \zeta(G_n) \geq \frac{(\ln 2)^2}{32} \ln \frac{200}{153} \frac{n}{(\ln n)^3} \right\} \rightarrow 1. \quad (0.1)$$

The construction uses classes of the four sizes $\alpha-2, \dots, \alpha-5$, where α is the usual independence-number phase. Signing each class as independent or complete moves its first-moment root down by order $n/(\ln n)^3$, uniformly through the whole phase cycle. An exact sign-summed overlap identity reduces the second moment to a bipartite configuration model. We sum partial diagonals, unequal-type dense containments, all intermediate high cells, and every residual even-subgraph attachment with logarithmic cost $o(n/(\ln n)^4)$. Paley–Zygmund gives a possibly rare cocolouring, and a proved Alon–Scott-type leftover argument amplifies it to high probability without losing the root separation.

The divergence question was posed by Erdős and Gimbel (1993, p. 263) and restated by Gimbel (2016, Section 7.4, p. 100). For historical comparison, the first-order dense-random-graph chromatic asymptotic is due to Bollobás (1988, Theorem 4); Janson, Łuczak, and Ruciński (2000, Section 7.4, Theorem 7.14 and Remark 7.15) give a textbook treatment. These references provide context only: the proof of (0.1) below does not import their results.

1. Notation and elementary facts

All logarithms are natural unless a base is displayed. Put

$$q = \ln 2, \quad N = \ln n, \quad w = \ln N. \quad (1.1)$$

For an integer s , let

$$\mu_s(v) = \binom{v}{s} 2^{-\binom{s}{2}}, \quad \mu_s = \mu_s(n). \quad (1.2)$$

We use the following standard inequalities in their stated forms.

1. For integers $m \geq 1$,

$$\ln(m!) = m \ln m - m + \frac{1}{2} \ln(2\pi m) + O(1/m). \quad (1.3)$$

In particular, uniformly for $m \geq 0$, with $0 \leq \ln 0 = 0$, the last two terms may be replaced by $O(\ln(m+1))$.

2. If a random variable Y is a function of r independent blocks and changing one block changes Y by at most one, then

$$\Pr\{|Y - \mathbb{E}Y| \geq t\} \leq 2 \exp(-2t^2/r). \quad (1.4)$$

We will use the corresponding one-sided bounds as well.

3. If $Z \geq 0$ and $0 < \mathbb{E}Z < \infty$, then

$$\Pr\{Z > 0\} \geq \frac{(\mathbb{E}Z)^2}{\mathbb{E}Z^2}. \quad (1.5)$$

4. If $X \sim \text{Bin}(m, 1/2)$, then

$$\Pr\{X \leq m/4\} \leq e^{-m/16}. \quad (1.6)$$

Indeed, exponential Markov with $t = \ln 3$ bounds the probability by $[3^{-(1/4)}(2/3)]^m = e^{-m/16}$.

5. For every nonnegative random variable X and $a > 0$, $P(X \geq a) \leq E X/a$.

The first is Stirling's estimate, the second is McDiarmid's bounded-differences inequality, and the third is the zero-threshold case of Paley–Zygmund. The fourth is the only binomial-tail estimate used below; the fifth is Markov's inequality. The bounded-differences formulation is the standard one recorded by McDiarmid (1989, pp. 148–188).

2. The complete independence-number phase

Define

$$\alpha_0 = 2 \log_2 n - 2 \log_2 \log_2 n + 2 \log_2(e/2) + 1, \quad \alpha = \lfloor \alpha_0 \rfloor, \quad \delta = \alpha_0 - \alpha. \quad (2.1)$$

Thus $0 \leq \delta < 1$. The estimates below are uniform on this entire interval, including sequences approaching either endpoint.

Lemma 2.1 (phase expansion)

There is a bounded continuous function $K(\delta)$ such that

$$\ln \mu_\alpha = \delta N + \left(\frac{2}{q} - \frac{1}{2} - \delta \right) w + K(\delta) + O(w^2/N). \quad (2.2)$$

In particular,

$$\mu_{\alpha+2} = n^{\delta-2+o(1)} \leq n^{-1+o(1)} = o(1), \quad (2.3)$$

and, with $a = \alpha - 2$,

$$\mu_a \geq cn^2 N^{2/q-5/2} \quad (2.4)$$

for an absolute $c > 0$ and all sufficiently large n .

Proof

Write

$$C = 1 + \ln q - q, \quad S = N - w + C, \quad b = 1 - \delta.$$

Then $\alpha = 2S/q + b$. Since $\alpha = O(N)$, (1.3) gives

$$\ln(n)_\alpha = \alpha N + \sum_{j=0}^{\alpha-1} \ln(1 - j/n) = \alpha N + O(\alpha^2/n),$$

and hence

$$\ln \mu_\alpha = \alpha \left(N - \ln \alpha + 1 - \frac{q}{2}(\alpha - 1) \right) - \frac{1}{2} \ln(2\pi\alpha) + O(1/N). \quad (2.5)$$

Expanding $\ln \alpha$ at $2N/q$, uniformly in b in $(0, 1]$, gives

$$N - \ln \alpha + 1 - \frac{q}{2}(\alpha - 1) = \frac{q\delta}{2} + \frac{w - C - bq/2}{N} + O(w^2/N^2). \quad (2.6)$$

Multiplication by α proves (2.2). For reference, the bounded constant is

$$\begin{aligned} K(\delta) &= \delta C + \frac{q}{2}\delta(1 - \delta) - \frac{2C}{q} - (1 - \delta) \\ &\quad - \frac{1}{2}\ln(2\pi) - \frac{q}{2} + \frac{1}{2}\ln q. \end{aligned} \quad (2.7)$$

The exact adjacent-size ratios are

$$\frac{\mu_{s+1}}{\mu_s} = \frac{n-s}{s+1}2^{-s}, \quad \frac{\mu_{s-1}}{\mu_s} = \frac{s}{n-s+1}2^{s-1}. \quad (2.8)$$

At $s=\alpha+O(1)$, the first is $\Theta(N/n)$ and the second is $\Theta(n/N)$, uniformly in δ . Applying the first ratio twice to (2.2) proves (2.3). More explicitly, $\delta(N - w) \geq 0$, while K is bounded, so (2.2) gives $\ln \mu_\alpha \geq (2/q - 1/2)w + \inf_\delta K(\delta) + o(1)$ uniformly in the phase. Hence $\mu_\alpha \geq c_1 N^{2/q-1/2}$. Applying the second ratio twice proves (2.4). $\square$

It follows at once from (2.3) and Markov's inequality that

$$\Pr\{\alpha(G_n) > \alpha + 1\} = o(1). \quad (2.9)$$

3. Continuous profile roots

For a class size u , put

$$d_u = 2^{\binom{u}{2}} u!.$$

Write $u=\alpha-i$, so that i is the deficit from α . Let

$$S_+ = \{-1, 0, 1, 2, \dots\}, \quad S_4 = \{2, 3, 4, 5\}. \quad (3.1)$$

At finite n , S_+ is cut off at $i=\alpha-1$, since class sizes are positive. For a real class count k , define

$$L_k = nN - n + k - \sum_i k_i \ln(k_i d_{\alpha-i}), \quad (3.2)$$

and let $L_S(n, k)$ be its maximum over real $k_i \geq 0$ satisfying

$$\sum_{i \in S} k_i = k, \quad \sum_{i \in S} (\alpha - i) k_i = n. \quad (3.3)$$

Set

$$T = \alpha - \frac{n}{k}. \quad (3.4)$$

The following lemma supplies all root geometry needed later.

Lemma 3.1 (root phase, derivative, and support comparison)

Let

$$s_0 = \alpha_0 - 1 - \frac{2}{q}, \quad T_0 = \alpha - s_0 = 1 + \frac{2}{q} - \delta. \quad (3.5)$$

For $S \in \{S_+, S_4\}$, and uniformly for $0 \leq c \leq q$, the equation $L_S(n, k) + ck = 0$ has a unique zero $r_{S,c}$ in the root corridor described below, and

$$\frac{n}{r_{S,c}} = s_0 + O(w/N), \quad T = \alpha - \frac{n}{r_{S,c}} = T_0 + O(w/N). \quad (3.6)$$

For every fixed $A > 0$, uniformly when $|n/k - s_0| \leq Aw/N$,

$$\frac{\partial}{\partial k} \{L_S(n, k) + ck\} = \frac{2}{q} N^2 + O(Nw). \quad (3.7)$$

For two supports R, S , at the same feasible k ,

$$\frac{L_R(n, k) - L_S(n, k)}{k} = \mathcal{F}_{n,R}(T) - \mathcal{F}_{n,S}(T), \quad (3.8)$$

where

$$\mathcal{F}_{n,S}(T) = \max_{\substack{\sum p_i = 1 \\ \sum i p_i = T}} \left[-\sum_i p_i \ln p_i + \sum_i p_i h_n(i) \right]. \quad (3.8a)$$

To make the uniformity precise, fix

$$K_* = [2/q - 1/10, 1 + 2/q + 1/10] \subset (2, 5). \quad (3.9a)$$

Then, uniformly for $T \in K_*$,

$$\mathcal{F}_{n,S}(T) \rightarrow \mathcal{F}_S(T) := \max_{\substack{\sum p_i = 1 \\ \sum i p_i = T}} \left[-\sum p_i \ln p_i - \frac{q}{2} \sum i^2 p_i \right]. \quad (3.9)$$

The effective Lagrange tilts are uniformly bounded on K_* . For S_4 , if $p_{n,i}(T)$ and $p_i(T)$ denote the finite and limiting optimizers, then

$$\sup_{T \in K_*} \max_{2 \leq i \leq 5} |p_{n,i}(T) - p_i(T)| = o(1), \quad \inf_{T \in K_*} \min_{2 \leq i \leq 5} p_{n,i}(T) > 0. \quad (3.9b)$$

All convergence and error terms are uniform in δ .

Proof

Put $s = n/k$ and $p_i = k_i/k$. Dividing (3.2) by k gives

$$\frac{L_S(n, n/s)}{n/s} = (s-1)N - s + 1 + \ln s + \max_p \left[-\sum p_i \ln p_i - \sum p_i \ln d_{\alpha-i} \right]. \quad (3.10)$$

There is an exact affine-plus-curved decomposition

$$-\ln d_{\alpha-i} = A_n + B_n i + h_n(i), \quad (3.11)$$

where

$$A_n = -\ln d_\alpha, \quad B_n = q\alpha - q/2 + \ln \alpha, \quad (3.12)$$

and, for $i \geq 0$,

$$h_n(i) = -\frac{q}{2}i^2 + \sum_{r=0}^{i-1} \ln(1 - r/\alpha). \quad (3.13)$$

For $i = -1$, direct subtraction gives $h_n(-1) = -q/2 + \ln(\alpha/(\alpha+1))$. Thus, for fixed i ,

$$h_n(i) = -qi^2/2 + O(i^2/\alpha), \quad (3.14)$$

while (3.13) also supplies the uniform bound $h_n(i) \leq -qi^2/2$. Since the target mean T_0 stays in the compact interval

$$\frac{2}{q} \leq T_0 \leq 1 + \frac{2}{q}, \quad (3.15)$$

the mean maps at two fixed tilts bracket all of K_* , for both supports and all sufficiently large n . The effective Lagrange tilt is therefore in a fixed compact interval. Its optimizer is proportional to $\exp(\lambda i + h_n(i))$. On every bounded tilt interval, the weights and their first moments are dominated by a constant multiple of $\exp(C|i| - qi^2/2)$, a summable sequence. Hence the partition functions and their first derivatives converge uniformly. Their mean maps are strictly increasing, with derivative equal to a positive variance; compactness then gives uniform convergence of the inverse tilts. This proves (3.9), and on the finite support S_4 also proves (3.9b). The affine terms in (3.11) prove the exact difference identity (3.8).

Substitute (3.11) into (3.10). If $T = \alpha - s$, the scalar part at $s = s_0$ is, by Stirling,

$$s_0 \left(N - \ln \alpha - \frac{q}{2}s_0 + \frac{q}{2} \right) - N + T_0 + 1 + \ln s_0 + \frac{q}{2}T_0^2 - \frac{1}{2} \ln(2\pi\alpha) + O(1/N). \quad (3.15a)$$

Now $q s_0/2 = N - w + \ln q - q$ exactly, while $\ln \alpha = \ln s_0 + T_0/s_0 + O(1/N^2)$ and $\ln s_0 = w + q - \ln q + O(w/N)$. These relations reduce (3.15a) to $O(w)$, uniformly in the phase. The entropy term is $O(1)$ by the Gaussian tail just proved. Hence

$$L_S(n, n/s_0)/(n/s_0) = O(w). \quad (3.16)$$

Write $\Psi_S(s) = L_S(n, n/s)/(n/s)$. Differentiating (3.10) by the envelope theorem, using the bounded effective tilt just proved and

$$B_n = 2N - w + O(1), \quad (3.17)$$

gives

$$\Psi'_S(s) = -N + O(w) \quad (3.18)$$

uniformly in a fixed neighbourhood of $\mathbf{s}_0$. For $\Psi_{S,c}(s) = \Psi_S(s) + c$, (3.16) gives $\Psi_{S,c}(s_0) = O(w)$ and (3.18) gives a negative derivative of magnitude at least $N/2$. Taking a sufficiently large fixed corridor constant brackets zero at $s = s_0 - Aw/N$ and $s = s_0 + Aw/N$, and strict monotonicity gives the unique corridor zero and (3.6), uniformly for $0 \leq c \leq q$.

Throughout any fixed corridor $|s - s_0| \leq Aw/N$, the same equations give $\Psi_{S,c}(s) = O_A(w)$. Since $L_S + ck = k\Psi_{S,c}(s)$ and $ds/dk = -s/k$,

$$\frac{d}{dk}\{L_S + ck\} = \Psi_{S,c}(s) - s\Psi'_S(s) = \frac{2}{q}N^2 + O_A(Nw). \quad (3.19)$$

This is (3.7). $\square$

4. A valid unrestricted lower location for χ

Let $\mathbf{r}_+(n) = \mathbf{r}_\{\mathbf{S}_+, 0\}$ be the zero in Lemma 3.1, and put

$$k_\chi^- := \lfloor r_+(n) \rfloor - \lceil N \rceil. \quad (4.1)$$

For an integer profile, direct enumeration gives

$$\mathbb{E}X_{\mathbf{k}} = \frac{n!}{\prod_i ((\alpha - i)!)^{k_i} k_i!} 2^{-\sum_i k_i \binom{\alpha-i}{2}}. \quad (4.2)$$

There are at most $(n+1)^{(\alpha+1)} = \exp(O(N^2))$ bounded profiles. Applying (1.3) to (4.2), uniformly even when some $k_i = 0$, yields

$$\ln E_{n,k,\alpha+1} \leq L_+(n, k) + O(N^2), \quad (4.3)$$

where the left side is the logarithm of the expected total number of unordered, exactly $\mathbf{k}$ -part, $(\alpha+1)$ -bounded colourings. By Lemma 3.1,

$$L_+(n, k_\chi^-) \leq -cN^3, \quad (4.4)$$

so the expectation in (4.3) is $o(1)$. On the event (2.9), any colouring with at most k_χ^- -nonempty parts can be refined, by splitting parts, to exactly k_χ^- -nonempty independent parts, all of size at most $\alpha+1$. Therefore

$$\Pr\{\chi(G_n) \leq k_\chi^-\} = o(1). \quad (4.5)$$

This is an unrestricted chromatic lower bound; the size cap has been removed probabilistically by (2.9).

5. The four-size signed first-moment advantage

The optimizer in (3.9) has the form

$$p_i = \frac{e^{\lambda i - q i^2 / 2}}{\sum_{j \in S} e^{\lambda j - q j^2 / 2}}, \quad (5.1)$$

where λ is chosen to give mean T . Define

$$D_4(\delta) = \mathcal{F}_{S_+}(T_0) - \mathcal{F}_{S_4}(T_0), \quad \gamma_4 = \ln \frac{200}{153}. \quad (5.2)$$

Lemma 5.1 (uniform entropy certificate)

For every $0 \leq \delta \leq 1$,

$$0 \leq D_4(\delta) < \ln \frac{153}{100}, \quad q - D_4(\delta) > \gamma_4. \quad (5.3)$$

Proof

Let λ_4 be the tilt for S_4 . At $\lambda = 2q$, shift the index by two. The four-size mean is no larger than

$$2 + \frac{\sum_{j \geq 0} j 2^{-j^2/2}}{\sum_{j \geq 0} 2^{-j^2/2}} < 2 + \frac{4}{5} < \frac{2}{q}. \quad (5.4)$$

For the first strict inequality, retain the first three terms of the denominator, sum the numerator through $j=3$, and bound its remaining geometric tail by $1/60$; the elementary bounds $2/3 < q < 5/7$ prove the second. At $\lambda = 9q/2$, put $t = 2^{-1/8}$. The four weights are proportional to

$$t^{25}, \quad t^9, \quad t, \quad t.$$

Their mean is greater than four because

$$t - t^9 - 2t^{25} = t(1 - t^8 - 2t^{24}) = t/4 > 0, \quad (5.5)$$

whereas $1 + 2/q < 4$. Hence

$$2q < \lambda_4 < 9q/2. \quad (5.6)$$

For a tilt λ , let $L(\lambda)$ be the total weight of deficits $-1, 0, 1$ divided by the weight of deficits $2, 3, 4, 5$, and let $H(\lambda)$ be the analogous ratio for deficits at least six. The first ratio decreases and the second increases with λ . Directly summing the displayed finite terms and bounding the Gaussian tails gives

$$L(2q) < \frac{51}{100}, \quad H(3q) < \frac{1}{50}, \quad L(3q) < \frac{3}{25}, \quad H(9q/2) < \frac{1}{4}. \quad (5.7)$$

Here are explicit checks. At $2q$, with $b=2^{-1/2}$,

$$L(2q) = \frac{b + 1/4 + b/16}{1 + b + 1/4 + b/16} < 51/100.$$

At $3q$, after centring at deficit three, the denominator is $5/4+2b$; the high numerator is at most $b/16+1/256+1/3968$, and the low numerator is $1/256+b/16+1/4$. The inequalities follow from $7/10 < b < 71/100$. Finally, after division by t , the tail above deficit five at $9q/2$ begins with t^8, t^{24}, t^{48} ; the rest is below $1/60$, and

$$3(t^8 + t^{24}) + 4/60 < 2, \quad (5.8)$$

which is equivalent to $H(9q/2) < 1/4$.

If $\lambda_4 \leq 3q$, monotonicity and (5.7) give $L+H < 53/100$; if $\lambda_4 \geq 3q$, they give $L+H < 37/100$. Evaluating the dual function for S_+ at the S_4 tilt yields

$$D_4 \leq \ln(1 + L(\lambda_4) + H(\lambda_4)) < \ln(153/100).$$

Since S_4 is a subset of S_+ , the loss is nonnegative. Subtracting from $q=\ln 2$ completes the proof. $\square$

For a fixed partition into k classes, assigning each class the declaration “independent” or “complete” multiplies its first moment by 2^k : every declaration prescribes all internal edge bits, and all 2^k declarations have probability $2^{-\sum \binom{u}{2}}$. Let $r_4^{\text{co}} = r_{\{S_4, q\}}$ be the unique corridor zero of

$$L_{S_4}(n, k) + qk. \quad (5.9)$$

At $k=r_+$, Lemma 3.1 and (5.2) give

$$L_{S_4}(n, r_+) + qr_+ = r_+ \{q - D_4(\delta) + o(1)\}. \quad (5.10)$$

Both roots lie in the corridor (3.6), so the whole interval between them is inside a fixed corridor. Since $r_+ = (q/2 + o(1))n/N$, the mean-value theorem and the strengthened estimate (3.7) give

$$r_+ - r_4^{\text{co}} = \left(\frac{q^2}{4} \{q - D_4(\delta)\} + o(1) \right) \frac{n}{N^3}. \quad (5.11)$$

In particular, for all sufficiently large n ,

$$r_+ - r_4^{\text{co}} \geq \frac{q^2 \gamma_4}{8} \frac{n}{N^3}. \quad (5.12)$$

Choose the midpoint integer

$$k_{\text{co}} = \left\lceil \frac{r_4^{\text{co}} + r_+}{2} \right\rceil. \quad (5.13)$$

At this exact k , let $p_i^*(n)$ be the exact finite- n maximizer of $L_{\{S_4\}}$. Its Lagrange equations give

$$p_i^{(n)} \propto d_{\alpha-i}^{-1} e^{\lambda_n^{\text{raw}} i} \propto e^{\hat{\lambda}_n i + h_n(i)}, \quad \hat{\lambda}_n = B_n + \lambda_n^{\text{raw}}. \quad (5.14)$$

subject to mean $\alpha - n/k_{co}$. Lemma 3.1 makes these proportions converge uniformly to (5.1), with $\hat{\lambda}_n \rightarrow \lambda_4(T_0)$. Equation (5.6) also shows directly that the four limiting weights differ by at most a fixed factor. Thus

$$\min_{2 \leq i \leq 5} p_i^{(n)} \geq c_p > 0 \quad (5.15)$$

uniformly in the phase.

Round $k_{co} p_i^{(n)}$ to preliminary integers $\tilde{k}_i$. Define the signed constraint errors

$$e_0 = \sum_i \tilde{k}_i - k_{co}, \quad e_1 = \sum_i i \tilde{k}_i - (\alpha k_{co} - n),$$

so $e_0, e_1 = O(1)$, and add

$$\Delta k_2 = e_1 - 3e_0, \quad \Delta k_3 = 2e_0 - e_1. \quad (5.16)$$

Indeed, $\Delta k_2 + \Delta k_3 = -e_0$ and $2\Delta k_2 + 3\Delta k_3 = -e_1$. This enforces both constraints exactly, changes only $O(1)$ counts, and preserves positivity by (5.15). Because the displacement is tangent to both constraints and starts at the exact finite optimizer, its first variation vanishes and its Hessian cost is $O(1/k_{co})$. Applying (1.3) to the five factorials gives an exact integer vector

$$\mathbf{k} = (k_2, k_3, k_4, k_5), \quad u_i = \alpha - i, \quad k_i = \Theta(n/N), \quad (5.17)$$

whose signed first moment is

$$\mathbb{E} Z_{\mathbf{k}}^{\text{sgn}} = 2^{k_{co}} \frac{n!}{\prod_i (u_i!)^{k_i} k_i!} 2^{-\sum_i k_i \binom{u_i}{2}}, \quad (5.18)$$

and

$$\ln \mathbb{E} Z_{\mathbf{k}}^{\text{sgn}} = L_{S_4}(n, k_{co}) + q k_{co} + O(N) \geq c_Z k_{co} \quad (5.19)$$

for a phase-independent $c_Z > 0$. The last inequality follows from the midpoint distance, (3.7), and (5.12). Every u_i tends to infinity, so the independent and complete declarations of one part are disjoint. Thus $Z_{\{\mathbf{k}\}^{\text{sgn}}} > 0$ implies an actual cocolouring with k_{co} parts.

Finally, (4.1), (5.12), and (5.13) give

$$k_{\chi}^- - k_{co} \geq \left(\frac{q^2 \gamma_4}{16} - o(1) \right) \frac{n}{N^3}. \quad (5.20)$$

6. Exact signed second-moment representation

Temporarily label all slots within each of the four types. This multiplies the unordered witness by the deterministic factor $\prod_i k_i!$ and hence does not change its normalized second moment. Choose two independent uniform ordered partitions of the profile. If their slots are indexed by $\mathbf{a}$ and $\mathbf{b}$, put

$$r_{ab} = |V_a \cap W_b|. \quad (6.1)$$

The overlap matrix has the exact law

$$p(r) = \frac{\prod_a s_a! \prod_b t_b!}{n! \prod_{a,b} r_{ab}!}, \quad (6.2)$$

where here both degree lists are the multiset given by (5.17). Equivalently, place $\mathbf{s}_a$ stubs at each row slot and $\mathbf{t}_b$ stubs at each column slot and take a uniform perfect matching of the two n -stub sets.

Let $H(\mathbf{r})$ be the simple bipartite graph whose edges are cells with $r_{ab} \geq 2$, omitting isolated slot vertices. Write

$$W = \sum_{a,b} \binom{r_{ab}}{2}, \quad \beta(H) = |E(H)| - |V(H)| + c(H). \quad (6.3)$$

Lemma 6.1 (exact sign sum)

The normalized local factor of the pair of partitions is

$$A_\zeta(r) = 2^{W+c(H)-|V(H)|} = \left(\prod_{a,b} g(r_{ab}) \right) 2^{\beta(H)}, \quad (6.4)$$

where

$$g(0) = g(1) = g(2) = 1, \quad g(x) = 2^{\binom{x}{2}-1} \quad (x \geq 3). \quad (6.5)$$

Consequently

$$\frac{\mathbb{E}(Z_{\mathbf{k}}^{sgn})^2}{(\mathbb{E}Z_{\mathbf{k}}^{sgn})^2} = \sum_r p(r) A_\zeta(r). \quad (6.6)$$

Proof

Let $B = \sum_a \text{binom}(s_a, 2)$, which is the number of internal edge bits prescribed by one partition. A row sign and a column sign are compatible on a cell of size at least two exactly when they agree. Therefore signs must be constant on every component of H . There are

$$2^{2k_{co}-|V(H)|+c(H)}$$

compatible pairs of sign assignments, including the free signs on slots isolated from H . For each compatible pair, W edge bits are prescribed twice, so the probability of all constraints is

$2^{-(2B-W)}$. Division by the square of the one-partition signed probability $(2^{k_{co}} 2^{-B})^2$ gives the first expression in (6.4).

For every support edge, split $\text{binom}(r, 2)$ as $(\text{binom}(r, 2) - 1) + 1$. Since $|E| - |V| + c = \text{beta}$, this gives the second expression. Averaging over (6.2) proves (6.6). $\square$

The identity

$$2^{\beta(H)} = \#\{F \subseteq E(H) : \deg_F(v) \text{ is even for every } v\} \quad (6.7)$$

will be used repeatedly. It is the elementary fact that the even subgraphs form the binary cycle space of dimension $\text{beta}(H)$.

We also need one exact configuration-model estimate.

Lemma 6.2 (joint prescribed-cell bound)

In a bipartite configuration model of total degree m_0 , with row degrees d_a and column degrees d'_b , prescribe distinct cells D and demands $x_{ab} \geq 1$. Put

$$x = \sum_D x_{ab}, \quad D_a = \sum_b x_{ab}, \quad D'_b = \sum_a x_{ab}.$$

Then

$$\Pr\{r_{ab} \geq x_{ab} \ \forall ab \in D\} \leq \frac{\prod_a (d_a)_{D_a} \prod_b (d'_b)_{D'_b}}{(m_0)_x \prod_{ab \in D} x_{ab}!}. \quad (6.8)$$

In particular, with $\theta_{ab} = e^{-d_a} e^{-d'_b} / m_0$, the right side is at most

$$\prod_{ab \in D} \frac{\theta_{ab}^{x_{ab}}}{x_{ab}!}. \quad (6.9)$$

Proof

Choose the required row and column stubs and biject them inside each prescribed cell. This gives the numerator of (6.8); any fixed set of x paired stubs occurs with probability $1/(m_0)_x$. A union bound proves (6.8). Next use $(d)_r \leq d^r$ and

$$(m_0)_x = x! \binom{m_0}{x} \geq (m_0/e)^x, \quad (6.10)$$

which proves (6.9). Notice that (6.8) retains the single global falling factorial before (6.9) is taken. $\square$

7. Exact partial diagonals

For a selected common subprofile $\text{ell} = (\text{ell}_i)$, put

$$L = \sum_i \ell_i, \quad m = \sum_i u_i \ell_i, \quad M = \sum_i \binom{u_i}{2} \ell_i. \quad (7.1)$$

The signed first moment of such partial partitions occupying any m vertices of $[n]$ is

$$Y_\ell^{sgn} = 2^L \frac{n!}{(n-m)! \prod_i (u_i!)^{\ell_i} \ell_i!} 2^{-M}. \quad (7.2)$$

Summing the possible matchings between selected row and column slots gives the exact common-subprofile contribution

$$D(\ell) = \frac{\prod_i \binom{k_i}{\ell_i}^2}{Y_\ell^{sgn}}. \quad (7.3)$$

In particular, $D(0)=1$ and $D(k)=1/E_{Z_k^{sgn}}$. Adding one common block gives

$$\frac{D(\ell + e_i)}{D(\ell)} = \frac{(k_i - \ell_i)^2}{2(\ell_i + 1)\mu_{u_i}(n-m)}. \quad (7.4)$$

At the opposite endpoint, put $\mathbf{h}=\mathbf{k}-\mathbf{e}_i$, $\mathbf{v}=\sum_i u_i \mathbf{h}_i$, and let $Z_{\mathbf{h}}^{sgn}(\mathbf{v})$ be the signed first moment of a complete profile $\mathbf{h}$ on $\mathbf{v}$ vertices. Exact factorial cancellation gives

$$D(k-h) = \frac{B(h)}{E_{Z_k^{sgn}}}, \quad B(h) = \left(\prod_i \binom{k_i}{h_i} \right) Z_h^{sgn}(\mathbf{v}), \quad (7.5)$$

and

$$\frac{B(h + e_i)}{B(h)} = \frac{2(k_i - h_i)\mu_{u_i}(v + u_i)}{(h_i + 1)^2}. \quad (7.6)$$

Lemma 7.1 (all common subprofiles)

For the exact midpoint profile,

$$\sum_{0 \leq \ell_i \leq k_i} D(\ell) = 1 + o(1). \quad (7.7)$$

Proof: the empty corner

Let

$$\eta = \frac{w}{32N}, \quad \lambda_i = \frac{k_i^2}{2\mu_{u_i}(n)}, \quad \Lambda = \sum_i \lambda_i. \quad (7.8)$$

Equations (2.4), (2.8), and $k_i = \Theta(n/N)$ imply

$$\Lambda = O(N^{-(2/q-1/2)}) = o(1). \quad (7.9)$$

If the final selected mass is at most ηn , every intermediate mass m' in (7.4) satisfies

$$\frac{\mu_{u_i}(n)}{\mu_{u_i}(n-m')} \leq (1-2\eta)^{-u_i} \leq N^{3/8}. \quad (7.10)$$

Iterating (7.4), in any order, yields

$$D(\ell) \leq \prod_i \frac{(N^{3/8} \lambda_i)^{\ell_i}}{\ell_i!}.$$

Thus

$$1 \leq \sum_{m \leq \eta n} D(\ell) \leq \exp(N^{3/8} \Lambda) = 1 + o(1). \quad (7.11)$$

Proof: the central range

Write

$$\begin{aligned} p_i &= k_i/k_{co}, & y_i &= \ell_i/k_{co}, & Y &= \sum_i y_i, & I &= \sum_i i y_i, \\ z_i &= p_i - y_i, & R &= \sum_i z_i = 1 - Y, & I_r &= \sum_i i z_i = T - I. \end{aligned} \quad (7.12)$$

The residual vertex fraction is

$$\rho = \frac{n - m}{n} = R + \frac{I - TY}{\alpha - T}. \quad (7.13)$$

Uniform Stirling estimates in (7.3), retaining zeros, give

$$\begin{aligned} \ln D(\ell) &= n \rho \ln \rho \\ &+ k_{co} \sum_i \{2p_i \ln p_i - 2(p_i - y_i) \ln(p_i - y_i) - y_i \ln y_i - y_i + y_i E_i\} + O(N), \end{aligned} \quad (7.14)$$

where

$$E_i = \ln k_{co} + \ln(u_i!) + u_i - u_i N + q \binom{u_i}{2} - q. \quad (7.15)$$

Exact subtraction and the definition of **alpha_0** give

$$E_{i+1} - E_i = N - \ln(\alpha - i) - q(\alpha - i) + q - 1 = -q\alpha/2 + O(1). \quad (7.16)$$

Let $\bar{E} = \sum p_i E_i$. Applying Stirling to the complete moment (5.18) gives

$$-\frac{1}{k_{co}} \ln \mathbb{E} Z_k^{sgn} = \sum_i p_i \ln p_i - 1 + \bar{E} + o(1). \quad (7.17)$$

The positive margin (5.19) bounds $\bar{E}$ above. Since $\sum p_i (i - T) = 0$, (7.16) implies

$$\sum_i y_i E_i \leq \frac{q\alpha}{2} (TY - I) + O(Y). \quad (7.18)$$

The remaining entropy expression in (7.14) is $O(Y \ln(e/Y))$, uniformly because every p_i is bounded below. If $\rho \geq 1/32$, then $R \geq 1/64$ for large n , and (7.13) gives

$$n\rho \ln \rho = k_{co}\alpha R \ln R + O(k_{co}Y). \quad (7.19)$$

Consequently

$$\ln D(\ell) \leq k_{co}\alpha \Phi_T(z) + Ck_{co}Y \ln(e/Y) + CN, \quad (7.20)$$

where

$$\Phi_T(z) = R \ln R + \frac{q}{2}(I_r - TR). \quad (7.21)$$

This rate is uniformly negative away from its two corners. Indeed,

$$I_r - TR \leq (5 - T)R, \quad I_r - TR = \sum_i (T - i)y_i \leq (T - 2)(1 - R), \quad (7.22)$$

and hence

$$\Phi_T \leq R\{\ln R + q(5 - T)/2\}, \quad \Phi_T \leq R \ln R + q(T - 2)(1 - R)/2. \quad (7.23)$$

On $1/64 \leq R \leq 47/100$, the first bound is at most $-Y/5000$, using $q < 0.6932$; on $47/100 \leq R \leq 1$, convexity of

$$R \ln R + (1 - q/2 + 1/200)(1 - R)$$

between $47/100$ and 1 gives the same conclusion, in fact with $1/200$ in place of $1/5000$. Thus, whenever

$$m > \eta n, \quad n - m > n/32, \quad (7.24)$$

we have $Y \geq \eta/2$. The leading term is at most $-k_{co}\alpha Y/5000$, whereas $\ln(e/Y) = O(w)$, $w/\alpha = o(1)$, and $N = o(k_{co}\alpha\eta)$. It therefore dominates both error terms in (7.20). Uniformly in this range,

$$D(\ell) \leq e^{-ck_{co}w}. \quad (7.25)$$

There are only $(k_{co}+1)^4$ vectors, so their total is $o(1)$.

Proof: the full corner

It remains to take $v = n - m \leq n/32$. Uniformly for the four sizes, $\mu_{\{u_i\}}(n) \leq n^{6+o(1)}$ by (2.2) and (2.8), while

$$\mu_{u_i}(v) \leq \mu_{u_i}(n)(v/n)^{u_i} \leq n^{-4+o(1)}. \quad (7.26)$$

Here $u_i = (2/q + o(1))N$ and $32^{-u_i} = n^{-10+o(1)}$. In any order of adding residual blocks, (7.6) is therefore below one for all sufficiently large n ; hence $B(h) \leq 1$. Equations (5.19) and (7.5) give

$$D(k - h) \leq e^{-c_Z k_{co}}. \quad (7.27)$$

Summing $(k_{co} + 1)^4$ residual vectors proves an $o(1)$ contribution. Together with (7.11) and (7.25), this proves (7.7). $\square$

8. Canonical high cells and dense endpoint transportation

Put

$$U = \alpha - 2, \quad R_0 = \lfloor U/2 \rfloor. \quad (8.1)$$

In every overlap matrix, the cells of multiplicity greater than R_0 form a matching: two such cells in one row or column would use more than U stubs. Let this canonical matching be

$$M = \{a_r b_r : 1 \leq r \leq h\}, \quad r_{a_r b_r} = j_r > R_0, \quad J = \sum_r j_r. \quad (8.2)$$

Selecting the exact stub pairs in these cells has incidence

$$\pi(M, j) = \frac{\prod_r (s_{a_r})_{j_r} (t_{b_r})_{j_r}}{(n)_J \prod_r j_r!}. \quad (8.3)$$

Conditional on the exposed pairs, the remaining matching is a uniform configuration model with total degree $n - J$ and the residual row and column degrees. It is restricted to put no further pair in a cell of M and no cell above R_0 . Multiplication of (8.3) by the exact residual contingency-table law cancels to (6.2). Thus every overlap matrix occurs exactly once.

We first sum the bare high matching, postponing all local and topological factors in the capped residual model. Index the four sizes as

$$u_i = a - i, \quad 0 \leq i \leq 3, \quad a = \alpha - 2. \quad (8.4)$$

For a typed full-containment matrix $L = (\ell_{ij})$, put

$$r_i = \sum_j \ell_{ij}, \quad c_j = \sum_i \ell_{ij}, \quad x_{ij} = \min(u_i, u_j), \quad J(L) = \sum_{ij} x_{ij} \ell_{ij}. \quad (8.5)$$

Its exact incidence, including its local signed rewards but not residual attachments, is

$$W(L) = \frac{\prod_i (k_i)_{r_i} \prod_j (k_j)_{c_j}}{\prod_{ij} \ell_{ij}!} \frac{1}{(n)_{J(L)}} \prod_{ij} \left[\frac{(u_i)_{x_{ij}} (u_j)_{x_{ij}}}{x_{ij}!} g(x_{ij}) \right]^{\ell_{ij}}. \quad (8.6)$$

For a margin vector $\mathbf{r}$, define

$$D(r) = \frac{\prod_i (k_i)_{r_i}^2 \prod_i [u_i! g(u_i)]^{r_i}}{\prod_i r_i! (n)_{m_r}}, \quad m_r = \sum_i u_i r_i. \quad (8.7)$$

This is exactly the common-subprofile weight (7.3).

Lemma 8.1 (geometric-mean transportation comparison)

For every feasible L ,

$$W(L) \leq \sqrt{D(r)D(c)} \frac{\sqrt{\prod_i r_i! c_i!}}{\prod_{ij} \ell_{ij}!} \prod_{ij} Q_{ij}^{\ell_{ij}}, \quad (8.8)$$

where $Q_{ii}=1$ and, if the two sizes are $t=s+d$, $1 \leq d \leq 3$,

$$Q_{ij} = (n+1)^{d/2} \frac{\sqrt{\binom{t}{d}}}{d!} 2^{-\{ds + \binom{d}{2}\}/2} \leq \frac{\eta_n^d}{d!}, \quad (8.9)$$

with

$$\eta_n = 2^{3/2} \sqrt{\frac{(n+1)a}{2^a}} = O(N^{3/2}/\sqrt{n}) = o(1). \quad (8.10)$$

Proof

For integer x , let $f(x) = \ln(n)_x$ and interpolate linearly between successive integers. Its successive slopes are $\ln(n-x)$, which decrease; therefore f is concave, and every slope is at most $\ln(n+1)$. Moreover,

$$\frac{m_r + m_c}{2} = J(L) + \frac{1}{2} \sum_{ij} |i-j| \ell_{ij}. \quad (8.11)$$

Concavity and $f'(x) \leq \ln(n+1)$ give

$$\frac{\sqrt{(n)_{m_r} (n)_{m_c}}}{(n)_{J(L)}} \leq (n+1)^{\frac{1}{2} \sum_{ij} |i-j| \ell_{ij}}. \quad (8.12)$$

For $t=s+d$, put $b_u = u! g(u)$. A direct calculation gives

$$\frac{\binom{t}{s} (s)_s}{s!} g(s) = b_s \binom{t}{d}, \quad \frac{b_t}{b_s} = \binom{t}{d} 2^{ds + \binom{d}{2}}. \quad (8.13)$$

Thus the local factor divided by $\sqrt{b_{sb_t}}$ is

$$\frac{\sqrt{\binom{t}{d}}}{d!} 2^{-\{ds + \binom{d}{2}\}/2}. \quad (8.14)$$

Dividing (8.6) by $\sqrt{D(r)D(c)}$, then using (8.12) and (8.14), proves (8.8)–(8.9). Finally $t \leq a$, $s \geq a-3$, and

$$2^a = \Theta(n^2/N^2) \quad (8.15)$$

uniformly in the phase. This proves (8.10). $\square$

Lemma 8.2 (sum of all endpoint tables)

$$\sum_L W(L) \leq \exp\{O(\sqrt{nN}) + O(N)\} \sum_r D(r) = \exp\{O(\sqrt{nN}) + O(N)\}. \quad (8.16)$$

Proof

For fixed margins, write

$$A_L = \frac{\prod_i r_i!}{\prod_{ij} \ell_{ij}!}, \quad C_L = \frac{\prod_j c_j!}{\prod_{ij} \ell_{ij}!}. \quad (8.17)$$

Cauchy's inequality gives

$$\sum_L \sqrt{A_L C_L} Q^L \leq \left(\sum_L A_L Q^L \right)^{1/2} \left(\sum_L C_L Q^L \right)^{1/2}. \quad (8.18)$$

Dropping the column constraints in the first sum and the row constraints in the second produces two exact multinomial expansions:

$$\sum_L A_L Q^L \leq \prod_i \left(\sum_j Q_{ij} \right)^{r_i}, \quad \sum_L C_L Q^L \leq \prod_j \left(\sum_i Q_{ij} \right)^{c_j}. \quad (8.19)$$

Every row and column sum of Q is at most $1+C$ `eta_n`, and the total margin is at most k_{co} . Hence the fixed-margin cost is $\exp(O(\text{eta_n } k_{co}))$. There are only $O(k_{co}^4)$ margin vectors, and

$$\left(\sum_r \sqrt{D(r)} \right)^2 \leq O(k_{co}^4) \sum_r D(r). \quad (8.20)$$

Use Lemma 7.1 and `eta_n k_co=O(sqrt(nN))` to finish. $\square$

Lemma 8.3 (all nonendpoint high multiplicities)

The full-containment sum (8.16) remains $\exp(o(n/N^4))$ after all high-cell multiplicities $R_{0 < j < \min(u_i, u_j)}$ are inserted.

Proof

Let the smaller slot size be m , the larger $m+d$, with $0 \leq d \leq 3$. Replacing the endpoint multiplicity m by $m-e$ has exact local ratio

$$R_{m,d}(e) = \frac{\binom{m}{e}}{(d+1) \cdots (d+e)} 2^{-em+e(e+1)/2}. \quad (8.21)$$

For many decorated cells, with total deficit Q , the ratio of the one global denominators is

$$\frac{\binom{n}{J_0}}{\binom{n}{J_0-Q}} = (n - J_0 + Q)_Q \leq n^Q. \quad (8.22)$$

For $1 \leq e \leq m/4$, the consecutive ratio of the summands $n^e R_{m,d}(e)$ is

$$\frac{n(m-e)2^{-m+e+1}}{(e+1)(d+e+1)}. \quad (8.23)$$

The ratio of (8.23) at $e+1$ to its value at e is

$$2 \frac{m-e-1}{m-e} \frac{e+1}{e+2} \frac{d+e+1}{d+e+2}, \quad (8.24)$$

which increases for $e \leq m/4$ and $d \leq 3$. Here is a direct check of the one-turn assertion. If $r(e)$ denotes (8.23), interpolated for real e , then

$$\frac{d^2}{de^2} \ln r(e) = -\frac{1}{(m-e)^2} + \frac{1}{(e+1)^2} + \frac{1}{(d+e+1)^2} > 0 \quad (8.24a)$$

for $0 \leq e \leq m/4$ and all sufficiently large m . Thus the consecutive ratios are log-convex and their maximum is at an endpoint. At $e = 0$ the ratio is $O(N^3/n)$, by (8.15), and at $e = \lfloor m/4 \rfloor$ it is $n^{-1/2+o(1)}$. The series therefore decreases geometrically and

$$\sum_{1 \leq e \leq m/4} n^e R_{m,d}(e) = O(N^3/n). \quad (8.25)$$

The passage from this per-cell estimate to the full skeleton sum is as follows. For a full-containment table L , temporarily distinguish its endpoint cells and let $\mathcal{S}(L)$ be all near-containment skeletons obtained by assigning to each such cell either $e = 0$ or $1 \leq e < m/4$. If $w(S)$ is its bare incidence and local weight, (8.21)–(8.25) give the nonnegative bound

$$\sum_{S \in \mathcal{S}(L)} w(S) \leq W(L) \prod_{c \in L} \left(1 + \sum_{1 \leq e < m_c/4} n^e R_{m_c, d_c}(e) \right). \quad (8.25a)$$

Distinguishing and then forgetting identical typed cells is exactly the multinomial expansion, so there is no additional multiplicity. Since a high skeleton is a matching and has at most k_{co} cells, (8.25a) is at most $W(L)$ times

$$(1 + O(N^3/n))^{k_{co}} = \exp(O(N^2)). \quad (8.26)$$

It remains to cover the middle strip, rather than silently importing an equitable-profile estimate. First expose a near-containment skeleton and let m_0 be its residual stub mass. Every unexposed high cell is then in the range $R_0 < j \leq 3a/4 + O(1)$. Let $E_{mid}(S)$ be the conditional bare middle-extension factor. If $m_0 \geq n/N^6$, expand over distinct residual cells and their threshold demands. With $\theta_{ab} = ed_a d'_b / m_0 \leq ea^2 / m_0$, Lemma 6.2 is applied jointly before the constraints are dropped, and gives

$$\begin{aligned}
E_{mid}(S) &\leq \sum_{\substack{D \text{ distinct residual cells} \\ R_0 < j_{ab} \leq 3a/4 + O(1)}} \prod_{ab \in D} g(j_{ab}) \Pr\{r_{ab} \geq j_{ab} \ \forall ab \in D\} \\
&\leq \prod_{ab} \left(1 + \sum_{R_0 < j \leq 3a/4 + O(1)} g(j) \frac{\theta_{ab}^j}{j!} \right) \leq \exp(\Xi_4),
\end{aligned} \tag{8.26a}$$

where

$$\Xi_4 \leq k_{co}^2 \sum_{a/2 < j \leq 3a/4 + O(1)} g(j) \frac{(ea^2/m_0)^j}{j!}. \tag{8.27}$$

Put $L = \log_2 n$ and $j = xL$. Uniformly for $1 + o(1) \leq x \leq 3/2 + o(1)$,

$$\begin{aligned}
\log_2 \left[k_{co}^2 g(j) \frac{(ea^2/m_0)^j}{j!} \right] &\leq 2L + \frac{j^2}{2} - j \log_2 m_0 + O(j \log_2 a) \\
&\leq -cL^2.
\end{aligned} \tag{8.28}$$

Indeed, the quadratic coefficient is $x^2/2 - x \leq -3/8 + o(1)$ on this interval. Hence $\Xi_4 = 2^{-\{-\Omega(L^2)\}}$.

If instead $m_0 < n/N^6$, leave the entire residual matching unexposed. Every residual degree is at most a , so every residual cell has $r_e \leq a$ and therefore $\binom{r_e}{2} \leq ((a-1)/2)r_e$. The near skeleton is a matching, so pointwise

$$\beta(M \cup H_{res}) \leq |E(H_{res})| \leq m_0/2, \quad \sum_e \binom{r_e}{2} \leq (a-1)m_0/2. \tag{8.29}$$

All remaining local and topological factors together are therefore at most $\exp(O(am_0)) = \exp(O(n/N^5))$. Since $g \geq 1$ and $\beta \geq 0$, summing the residual matching probability first gives the explicit conditional bound

$$E_{mid}(S) \leq \mathbb{E}_{res} \left[\prod_e g(r_e) 2^{\beta(S \cup H_{res})} \right] \leq \exp(Cam_0). \tag{8.29a}$$

Combining (8.16), (8.25a)–(8.26a), and the two residual-mass branches now gives the global inequality

$$\sum_{\text{all high skeletons}} \text{bare weight} \leq e^{O(N^2)} \left(\sum_L W(L) \right) \max\{e^{\Xi_4}, e^{Cn/N^5}\}. \tag{8.29b}$$

Therefore

$$\sum_{\text{all high skeletons}} \text{bare weight} \leq \exp\{O(\sqrt{nN}) + O(N^2) + O(n/N^5)\} = \exp\{o(n/N^4)\}. \tag{8.30}$$

□

9. All residual local and cycle attachments

Fix an actual canonical high skeleton (M, j) as in (8.2), and let $m_0 = n - J$ be the residual total degree. Its residual degrees are at most U . Define

$$\mathcal{A}(M, j) = \mathbb{E}_{res} \left[\left(\prod_e g(r'_e) \right) 2^{\beta(M \cup H_{res})} \mathbf{1}_{\mathcal{E}(M, j)} \right], \quad (9.1)$$

where $\mathcal{E}(M, j)$ imposes the residual cap $r'_e \leq R_0$ and forbids a further pair in a cell of M . By (6.4), (8.3), and the exact conditional law, the normalized moment is exactly

$$\sum_{(M, j)} \pi(M, j) \left(\prod_{e \in M} g(j_e) \right) \mathcal{A}(M, j). \quad (9.2)$$

Lemma 9.1 (uniform attachment bound)

Uniformly over every canonical high skeleton,

$$\mathcal{A}(M, j) \leq \exp\{o(n/N^4)\}. \quad (9.3)$$

More precisely, for an absolute C , it is at most $\exp(CN^8)$ when $m_0 \geq n/N^6$ and at most $\exp(Cn/N^5)$ otherwise. No lower bound on $\mathcal{A}$ is asserted or needed.

Proof: large residual degree

Suppose $m_0 \geq n/N^6$. Since

$$U = (2 + o(1)) \log_2 n, \quad \log_2 m_0 = \log_2 n - O(\log_2 N), \quad (9.4)$$

we have $U < 3 \log_2 m_0$ for all sufficiently large n . For a residual cell outside M , put

$$\theta_{ab} = \frac{ed_a d'_b}{m_0}, \quad \Delta_x = g(x) - g(x-1), \quad (9.5)$$

and

$$\lambda_{ab} = \sum_{x=3}^{R_0} \Delta_x \frac{\theta_{ab}^x}{x!}, \quad q_{ab} = \frac{\theta_{ab}^2}{2} + \lambda_{ab}. \quad (9.6)$$

Set both quantities to zero on M . Let $\theta_* = eU^2/m_0$. The ratio of consecutive majorants $g(x)\theta_*^x/x!$ is

$$\frac{2^x \theta_*}{x+1}, \quad (9.7)$$

and increases with x . Hence the sequence is log-convex. Its decreasing head is dominated by the $x=3$ term. Consecutive values of (9.7) grow by the factor $2(x+1)/(x+2) \geq 8/5$, so only $O(1)$ transition terms can have ratio between $1/2$ and 1 . Its increasing tail is dominated by its right endpoint, and

$$\log_2 \left(g(R_0) \frac{\theta_*^{R_0}}{R_0!} \right) \leq \frac{R_0^2}{2} - R_0 \log_2 m_0 + O(R_0 \log_2 U) = -\Omega(R_0 \log n), \quad (9.8)$$

using $R_0=U/2$ and (9.4). Thus, uniformly in the cell,

$$\lambda_{ab} \leq C\theta_{ab}^3, \quad q_{ab} \leq C\theta_{ab}^2. \quad (9.9)$$

Expand each capped local factor as

$$g(r'_e) = 1 + \sum_{x=3}^{R_0} \Delta_x \mathbf{1}_{\{r'_e \geq x\}}. \quad (9.10)$$

If an edge is selected in a fixed even subgraph, use instead

$$\mathbf{1}_{\{r'_e \geq 2\}} g(r'_e) = \mathbf{1}_{\{r'_e \geq 2\}} + \sum_{x=3}^{R_0} \Delta_x \mathbf{1}_{\{r'_e \geq x\}}. \quad (9.11)$$

The terms on the right of (9.11) are alternatives, so a triple cell is not charged twice. Fix an even subgraph in (6.7), expand (9.10)–(9.11), and apply Lemma 6.2 jointly to all demanded cells. Dropping the cap only after this nonnegative expansion gives

$$\mathcal{A}(M, j) \leq e^{\Lambda_0} \sum_{F \text{ even}} \prod_{e \in F \setminus M} q_e, \quad \Lambda_0 = \sum_e \lambda_e. \quad (9.12)$$

The degree sums give

$$\Lambda_0 \leq \frac{C}{m_0^3} \left(\sum_a d_a^3 \right) \left(\sum_b (d'_b)^3 \right) \leq C \frac{U^4}{m_0}, \quad (9.13)$$

and

$$\max_a \sum_b q_{ab} \vee \max_b \sum_a q_{ab} \leq C \frac{U^3}{m_0} =: \tau = o(1). \quad (9.14)$$

Every even graph is a union of edge-disjoint simple cycles. Choose one such decomposition deterministically; dropping edge-disjointness gives

$$\sum_{F \text{ even}} \prod_{e \in F \setminus M} q_e \leq \exp \left\{ \sum_{C \text{ simple cycle}} \prod_{e \in C \setminus M} q_e \right\}. \quad (9.15)$$

For cycles disjoint from M , mark a row start and drop simplicity and the closing constraint. The weighted row and column sums are at most τ , and a bipartite cycle has length at least four. Hence

$$\sum_{C: C \cap M = \emptyset} \prod_{e \in C} q_e \leq \frac{n\tau^4}{1 - \tau^2}. \quad (9.16)$$

For a cycle meeting M , mark and orient one matching edge. There are at most $2h$ choices. Cutting at all matching edges leaves nonempty residual walks. The residual-walk kernel has row-sum norm at most

$$b = \sum_{\ell \geq 1} \tau^\ell = \frac{\tau}{1 - \tau}, \quad (9.17)$$

while traversal of the matching is a partial permutation of norm one. Thus, if the relaxed cycle uses r matching edges, its total remaining weight is at most b^r ; there is no fresh factor h after the first marked edge. Therefore

$$\sum_{C: C \cap M \neq \emptyset} \prod_{e \in C \setminus M} q_e \leq 2h \sum_{r \geq 1} b^r \leq Ch\tau. \quad (9.18)$$

Every high cell uses more than $U/2$ stubs, so $h < 2n/U$. Combining (9.12)–(9.18),

$$\mathcal{A}(M, j) \leq \exp \left[C \left\{ \frac{U^4}{m_0} + n\tau^4 + h\tau \right\} \right] \leq \exp(C' N^8) \quad (9.19)$$

at the cutoff $m_0 \geq n/N^6$.

Proof: small residual degree

Suppose $m_0 < n/N^6$. Since M is a matching, it is a forest, and adding a residual support edge increases cycle rank by at most one. Hence

$$\beta(M \cup H_{res}) \leq |E(H_{res})| \leq m_0/2. \quad (9.20)$$

Also, pointwise,

$$\sum_e \binom{r'_e}{2} \leq \frac{U-1}{2} \sum_e r'_e = \frac{U-1}{2} m_0. \quad (9.21)$$

Equations (6.5), (9.20), and (9.21) bound the integrand in (9.1) by $2^{\{Um_0/2\}}$. Therefore

$$\mathcal{A}(M, j) \leq \exp(O(Um_0)) \leq \exp(Cn/N^5). \quad (9.22)$$

Both exponents in (9.19) and (9.22) are $o(n/N^4)$. $\square$

Proposition 9.2 (normalized signed second moment)

For the exact midpoint profile,

$$\Lambda_n := \ln \frac{\mathbb{E}(Z_{\mathbf{k}}^{sgn})^2}{(\mathbb{E} Z_{\mathbf{k}}^{sgn})^2} = o(n/N^4). \quad (9.23)$$

Proof

Equation (9.2) is an exact canonical decomposition. Lemma 8.3 sums all bare high skeletons by $\exp(o(n^{1/4}))$, including the empty skeleton, all dense endpoint tables, every near-containment decoration, and the two middle-strip regimes. Lemma 9.1 bounds the conditional residual factor by the same exponential scale uniformly over every skeleton. Multiplication proves the upper bound in (9.23). The normalized second moment is at least one by variance, so its logarithm is nonnegative and (9.23) follows. $\square$

10. Rare-event amplification

Proposition 9.2 and (1.5) give the seed

$$\Pr\{Z_{\mathbf{k}}^{sgn} > 0\} \geq e^{-\Lambda_n}. \quad (10.1)$$

The ordinary-colouring concentration argument motivating this amplification was communicated by Noga Alon and is presented by Scott (2008/2017, Theorem 1). Lemmas 10.1 and 10.2 prove the precise simultaneous-leftover and rare-seed forms required here.

By the observation after (5.19), this implies

$$\Pr\{\zeta(G_n) \leq k_{co}\} \geq e^{-\Lambda_n}. \quad (10.2)$$

We now prove the amplification needed to turn this possibly rare event into a typical one.

Lemma 10.1 (simultaneous leftover colouring)

There is an absolute C such that, with probability $1 - o(1)$, every $U \subseteq [n]$ satisfies

$$\chi(G_n[U]) \leq C \frac{|U|}{N} + n^{1/3}. \quad (10.3)$$

Proof

Let H be the complement of G_n . Put $u_0 = \lceil n^{1/4} \rceil$. For any fixed u_0 -set, (1.6) gives probability $\exp(-\Omega(u_0^2))$ that its H -edge density is below $1/4$. There are at most $\exp(u_0 N)$ such sets, so a union bound shows that, with probability $1 - o(1)$, every u_0 -set has density at least $1/4$. By averaging over its u_0 -subsets, the same is then true of every larger set.

Start with any vertex set of size at least $n^{1/3}$. While its current size is at least u_0 , choose a vertex of maximum H -degree and replace the current set by its H -neighbourhood there. Density at least $1/4$ implies that this neighbourhood has size at least one quarter of the current size, up to an additive one. If s_t is the current size after t choices, then

$$s_{t+1} \geq (s_t - 1)/4, \quad s_t \geq 4^{-t} s_0 - 1/3. \quad (10.3a)$$

The chosen vertices form a clique in H . Starting from any $\lceil n^{1/3} \rceil$ -subset, (10.3a) stays above $n^{1/4}$ for at least $\lfloor N/(13 \ln 4) \rfloor$ steps. Thus it constructs at least cN vertices for an absolute $c > 0$. They are an independent set in G_n .

For an arbitrary U , repeatedly remove such independent sets and give each a new colour until fewer than $n^{1/3}$ vertices remain; colour the rest singly. This proves (10.3) simultaneously for every U . $\square$

Lemma 10.2 (amplification from a seed)

Suppose deterministic $k_n, \Lambda_n \geq 0$ satisfy

$$\Pr\{\zeta(G_n) \leq k_n\} \geq e^{-\Lambda_n}. \quad (10.4)$$

There is a deterministic $\varepsilon_n = o(1)$, independent of k_n, Λ_n, r , such that, uniformly for every deterministic choice $r = r(n) > 0$,

$$\Pr\left\{\zeta(G_n) > k_n + C\left(\frac{\sqrt{n\Lambda_n} + \sqrt{nr}}{N} + n^{1/3} + 1\right)\right\} \leq e^{-r} + \varepsilon_n. \quad (10.5)$$

Proof

For an induced set W , let $\mathbf{zeta}(W) = \mathbf{zeta}(G_n[W])$, and define

$$S_k = \max\{|W| : \zeta(W) \leq k_n\}. \quad (10.6)$$

Expose the random graph in $n-1$ independent vertex blocks, where block v contains the edges from v to earlier vertices. Changing one block changes S_k by at most one: after deletion of the affected vertex, every feasible induced set loses at most one vertex and the two configurations agree on the remainder. Therefore (1.4) applies.

Since $S_k = n$ exactly when $\mathbf{zeta}(G_n) \leq k_n$, (10.4) and the upper one-sided bounded-differences tail imply

$$n - \mathbb{E}S_k \leq \sqrt{(n-1)\Lambda_n/2}. \quad (10.7)$$

The lower tail with radius $\sqrt{(n-1)r/2}$ gives, except with probability e^{-r} ,

$$n - S_k \leq \sqrt{(n-1)\Lambda_n/2} + \sqrt{(n-1)r/2}. \quad (10.8)$$

Choose a maximizing set W and put $U = [n] \setminus W$. Concatenating a k_n -part cocolouring of $G[W]$ with an ordinary colouring of $G[U]$ gives

$$\zeta(G_n) \leq k_n + \chi(G_n[U]). \quad (10.9)$$

Intersect (10.8) with the simultaneous event in Lemma 10.1 and apply (10.3). Its failure probability is the same deterministic $\varepsilon_n = o(1)$ for every choice of the three deterministic parameters. This proves (10.5). $\square$

Apply Lemma 10.2 to (10.2). Put

$$B_n = \frac{n}{N^4}, \quad r_n = \sqrt{B_n}. \quad (10.10)$$

Then $r_n \rightarrow \infty$, and Proposition 9.2 gives

$$\frac{\sqrt{n\Lambda_n}}{N} = o(n/N^3), \quad \frac{\sqrt{nr_n}}{N} = o(n/N^3), \quad n^{1/3} = o(n/N^3). \quad (10.11)$$

Define the deterministic sequence

$$a_n = C \left(\frac{\sqrt{n\Lambda_n} + \sqrt{nr_n}}{N} + n^{1/3} + 1 \right). \quad (10.12)$$

Then (10.11) and Lemma 10.2 give

$$a_n = o(n/N^3), \quad \Pr\{\zeta(G_n) > k_{co} + a_n\} \leq e^{-r_n} + \varepsilon_n \rightarrow 0. \quad (10.13)$$

11. Completion of the proof

Intersect $\{\chi(G_n) > k_\chi^-\}$, whose probability tends to one by (4.5), with $\{\zeta(G_n) \leq k_{co} + a_n\}$, whose probability tends to one by (10.13). No independence is required; a union bound suffices. Equation (5.20) then gives, with high probability,

$$\chi(G_n) - \zeta(G_n) \geq \left(\frac{q^2\gamma_4}{16} - o(1) \right) \frac{n}{N^3}. \quad (11.1)$$

For every sufficiently large n , the right side is at least

$$c_* \frac{n}{N^3}, \quad c_* := \frac{q^2\gamma_4}{32} = \frac{(\ln 2)^2}{32} \ln \frac{200}{153} > 0. \quad (11.2)$$

This proves (0.1). Since $n/(\ln n)^3$ tends to infinity, for every fixed M ,

$$\Pr\{\chi(G_n) - \zeta(G_n) \geq M\} \rightarrow 1. \quad (11.3)$$

Every estimate used above is uniform for δ in $[0,1)$ and for the exact tangent-rounded finite- n profile. The conclusion therefore holds along all sufficiently large integers, including sequences approaching either side of every independence-number jump. It is a high-probability statement, not a density-subset or subsequence result. ■

12. Dependency and provenance note

The proof is logically ordered as follows. The elementary phase expansion and exact first-moment variational calculation construct the midpoint profile and prove its complete signed first-moment margin. That margin and elementary independence-set estimates prove the full partial-diagonal sum. The partial-diagonal sum feeds the dense four-type transportation estimate. Independently, the exact sign sum and the configuration model prove the uniform residual-attachment estimate. These two estimates combine only after both are established, yielding the normalized second moment. Paley–Zygmund and the proved leftover amplification are applied last. Thus there is no circular use of a second-moment conclusion in the profile construction. In particular, no ordinary-colouring second-moment theorem is invoked at the signed midpoint, where the ordinary first moment is exponentially small.

For provenance only, the separate dossier files from which the arguments were consolidated are EXCEPTIONAL_REGIME.md, ALPHA_MINUS_TWO_ROUTE.md, FOUR_SIZE_PARTIAL_RATES.md, DENSE_FOUR_TYPE_MATCHING.md, SIGNED_PROFILE_OVERLAP.md, RESIDUAL_ATTACHMENT.md, and ALON_CONCENTRATION_EXTENSION.md. No step of the proof above depends on those cross-references. The cited historical sources confirm the definition, the original divergence question, and the standard first-order chromatic asymptotic; none supplies a step of the proof above.

13. References

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